

Chapter 7 — Swaps (Part 3)

J.C. Hull — *Options, Futures and Other Derivatives*

Structure of Part 3

This final part focuses on:

- 1. Risks in Swap Markets
- 2. Credit Risk in Swaps
- 3. Swap Market Applications
- 4. Comparative Advantage Criticism
- 5. Advanced Swap Structures
- 6. Practical Hedging Insights
- 7. Exam-Focused Summary
- 8. Master Formula Sheet

This section is extremely important because it connects swap theory with:

- Real financial markets
 - Risk management
 - Banking practice
 - Derivatives trading
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7.7 Risks in Swap Markets

Although swaps are highly useful financial instruments, they involve substantial risks.

The major risks are:

Risk Type	Description
Counterparty Risk	One party defaults

Risk Type	Description
Market Risk	Interest rates/exchange rates move adversely
Liquidity Risk	Difficult to unwind positions
Operational Risk	Settlement/documentation failures
Legal Risk	Contract enforceability issues
Funding Risk	Difficulty financing hedge positions

Counterparty Credit Risk

The most important risk in OTC swap markets is:

Counterparty Credit Risk

Because swaps are privately negotiated:

- No exchange guarantees performance
- One counterparty may fail to pay

This is fundamentally different from exchange-traded futures.

Nature of Credit Exposure

Swap exposure is unusual because:

Exposure changes over time.

At initiation:

$$V_{swap} \approx 0$$

But later:

- Interest rates change
- Exchange rates change
- Swap value changes

Thus one party may owe large amounts to another.

Positive and Negative Swap Values

If swap value becomes:

Positive

- Swap is an asset
- Counterparty owes money

Negative

- Swap is a liability
 - Party owes money to counterparty
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Example — Credit Exposure

Suppose:

- Bank enters interest rate swap
- Swap initially worth zero

After interest-rate movements:

$$V_{swap} = +5,000,000$$

Then:

- Bank has \$5 million exposure to counterparty.

If counterparty defaults:

- Bank may lose significant value.

Credit Risk in Currency Swaps

Currency swaps usually involve:

Larger credit exposure

Why?

Because:

- Principal amounts are exchanged
- Exchange rates fluctuate
- Long maturities are common

Thus market values can become very large.

Mitigating Credit Risk

Financial institutions reduce swap credit risk using:

1. Netting Agreements

Positive and negative positions offset.

2. Collateral Agreements

Counterparties post collateral.

3. Central Clearing

Some swaps now cleared through CCPs.

4. Credit Limits

Banks impose exposure limits.

5. Frequent Marking to Market

Swap values updated regularly.

Netting Agreements

Netting is one of the most important risk-reduction techniques.

Suppose:

- Bank A owes Bank B \$10 million on one swap
- Bank B owes Bank A \$8 million on another swap

Without netting:

- Gross exposure = \$18 million

With netting:

$$10 - 8 = 2$$

Net exposure:

2 million

Thus netting dramatically reduces risk.

Collateral in Swap Markets

Modern swap markets heavily use:

Margin and Collateral Agreements

Collateral may include:

- Cash
- Government securities
- High-quality bonds

Collateral reduces expected losses from default.

Central Clearing of Swaps

After the 2008 financial crisis:

Central clearing became much more important.

Clearinghouses:

- Stand between counterparties
 - Guarantee performance
 - Reduce systemic risk
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OTC vs Cleared Swaps

Feature	OTC Swap	Cleared Swap
Customization	High	Lower
Credit Risk	Higher	Lower
Liquidity	Lower	Higher
Regulation	Less	More

Market Risk in Swaps

Swap values fluctuate because:

- Interest rates change
- Yield curves shift
- Exchange rates move
- Credit spreads change

Thus swaps are exposed to:

Mark-to-Market Risk

Interest Rate Sensitivity

An interest rate swap behaves similarly to:

A bond portfolio

Therefore:

- Duration
- Convexity
- Yield curve exposure

are all important.

Duration of a Swap

Swaps possess:

Interest Rate Duration

Generally:

Position	Duration Exposure
Receive Fixed	Positive duration
Pay Fixed	Negative duration

Why?

Receiving fixed resembles:

Owning a fixed-rate bond

Thus:

- Swap gains value when rates fall

Paying fixed resembles:

Shorting a fixed-rate bond

Thus:

- Swap gains value when rates rise
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Liquidity Risk

Some swaps may be difficult to terminate before maturity.

Liquidity risk increases for:

- Long-dated swaps

- Customized structures
 - Exotic swaps
 - Illiquid currencies
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Operational Risk

Operational problems may include:

- Incorrect settlement
- Documentation mistakes
- System failures
- Human error

Large financial institutions devote enormous resources to managing these risks.

Legal Risk

Swap contracts require enforceable legal agreements.

Problems may arise due to:

- Jurisdiction differences
- Bankruptcy law
- Documentation ambiguity

Thus:

ISDA agreements are critical.

7.8 Criticism of Comparative Advantage Argument

Hull points out that:

Comparative advantage is not a complete explanation for swaps.

This is a very important conceptual insight.

Why Comparative Advantage Is Incomplete

Suppose:

- Markets are perfectly efficient
- No transaction costs exist

Then true arbitrage opportunities should disappear quickly.

Yet swaps continue to exist.

Therefore additional explanations are necessary.

More Realistic Explanations

Modern swap usage is often driven by:

1. Hedging Needs

Firms want to manage interest-rate exposure.

2. Asset-Liability Matching

Banks manage duration mismatch.

3. Regulatory Advantages

Capital regulations differ across markets.

4. Tax Considerations

Different financing structures have different tax treatment.

5. Market Segmentation

Some firms have easier access to certain markets.

6. Accounting Considerations

Swaps may improve accounting treatment.

Key Modern View

Today swaps are viewed primarily as:

Risk management tools

rather than simple comparative-advantage trades.

7.9 Applications of Swaps

Swaps are used extensively across financial markets.

Corporate Applications

Corporations use swaps to:

- Convert fixed debt to floating debt
 - Convert floating debt to fixed debt
 - Hedge exchange-rate risk
 - Reduce financing uncertainty
 - Match cash flows
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Banking Applications

Banks use swaps for:

Asset-Liability Management (ALM)

Banks often:

- Borrow short-term
- Lend long-term

This creates:

Duration mismatch

Swaps help reduce this mismatch.

Insurance Companies and Pension Funds

These institutions often prefer:

- Stable long-term cash flows

Thus they frequently:

- Receive fixed
 - Pay floating
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Hedge Funds

Hedge funds use swaps for:

- Speculation
 - Yield curve trading
 - Relative value strategies
 - Leveraged positions
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Governments and Sovereigns

Governments sometimes use swaps to:

- Manage foreign debt
 - Alter maturity structure
 - Hedge currency exposure
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7.10 Advanced Swap Structures

Over time, many specialized swaps developed.

Basis Swaps

A basis swap exchanges:

One floating rate for another floating rate

Example:

- LIBOR vs SOFR
 - 3-month LIBOR vs 6-month LIBOR
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Amortizing Swaps

In amortizing swaps:

Notional principal declines over time.

Useful for:

- Mortgage-related cash flows
 - Project financing
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Accreting Swaps

In accreting swaps:

Notional principal increases over time.

Forward Swaps

A forward swap begins:

At a future date

These are sometimes called:

Deferred swaps

Swaptions

A swaption is:

An option on a swap

Holder has right to:

- Enter swap in future
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Types of Swaptions

Payer Swaption

Right to:

- Pay fixed
 - Receive floating
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Receiver Swaption

Right to:

- Receive fixed

- Pay floating
-

Commodity Swaps

Commodity swaps exchange:

- Fixed commodity price
- Floating commodity price

Used heavily in:

- Oil markets
 - Energy markets
 - Metals markets
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Equity Swaps

Equity swaps exchange:

- Equity return
- Interest-rate cash flows

Example:

ParseError: KaTeX parse error: Expected 'EOF', got '&' at position 8: \text{S\&P 500 Return} \...

Credit Default Swaps (CDS)

One of the most important modern derivatives:

Credit Default Swap

A CDS provides insurance against:

Default risk

Protection buyer:

- Pays periodic premium

Protection seller:

- Pays compensation if default occurs
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Importance of CDS Market

CDS markets became extremely important before the:

2008 Financial Crisis

The crisis demonstrated:

- Interconnectedness of derivatives markets
 - Systemic risk in OTC derivatives
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Key Formulas from Part 3

Interest Rate Swap Value

Fixed Payer

$$V_{swap} = B_{floating} - B_{fixed}$$

Floating Payer

$$V_{swap} = B_{fixed} - B_{floating}$$

Fixed Bond Value

$$B_{fixed} = \sum_{i=1}^n K e^{-r_i t_i} + L e^{-r_n t_n}$$

Currency Swap Value

$$V_{swap} = B_{domestic} - S_0 B_{foreign}$$

Comparative Advantage Gain

$$\text{Difference in Fixed Spreads} - \text{Difference in Floating Spreads}$$

Swap Spread

$$\text{Swap Spread} = \text{Swap Rate} - \text{Treasury Yield}$$

Key Exam Concepts from Part 3

You must understand:

- Counterparty credit risk
- Netting and collateral
- Cleared vs OTC swaps
- Market risk in swaps
- Swap duration exposure
- Basis swaps
- Swaptions
- CDS
- Swap spreads
- Applications of swaps
- Comparative advantage criticism

Final Intuition of Part 3

Core ideas:

1. Swap markets are heavily exposed to counterparty credit risk.
2. Netting and collateral are essential risk-management tools.
3. Swaps are used extensively for hedging and balance-sheet management.
4. Modern swap markets are closely linked to banking system stability.
5. Advanced swap structures extend the flexibility of derivatives markets.
6. Swap spreads contain valuable information about financial stress.
7. Swaps are central to modern fixed-income and OTC derivatives markets.